

UNIT-I: LOGIC AND PROOF TECHNIQUES

5. PROOF TECHNIQUES AND MATHEMATICAL INDUCTION

5.1 Introduction to Mathematical Proofs

Mathematics is based on logical reasoning. A mathematical statement is accepted as true only after a valid proof has been provided.

A **proof** is a sequence of logically justified statements that establishes the truth of a mathematical proposition.

A proof begins with known facts, definitions, axioms, or previously established theorems and proceeds through logical reasoning to reach the desired conclusion.

Why Are Proofs Important?

Proofs help us:

1. Verify mathematical statements.
 2. Develop logical reasoning skills.
 3. Establish general results.
 4. Explain why a statement is true.
 5. Avoid incorrect conclusions.
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Example

Statement:

"The sum of two even integers is even."

Checking a few examples:

$$2 + 4 = 6$$

$$8 + 10 = 18$$

is not sufficient.

A proof must establish the result for **all** even integers.

5.2 Methods of Proof

The most common proof techniques are:

1. Direct Proof
 2. Proof by Contrapositive
 3. Proof by Contradiction
 4. Proof by Cases
 5. Biconditional Proof
 6. Existence Proof
 7. Uniqueness Proof
 8. Mathematical Induction
 9. Strong Mathematical Induction
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5.3 Direct Proof

Definition

In a direct proof, we assume the hypothesis is true and use definitions, axioms, and logical reasoning to derive the conclusion.

General Structure

Given:

If p , then q

Direct proof:

Step 1: Assume p .

Step 2: Use logical reasoning.

Step 3: Show q .

Example 1

Theorem

The sum of two even integers is even.

Proof

Let

$$a = 2m$$

and

$$b = 2n$$

where m and n are integers.

Then

$$a + b = 2m + 2n$$

$$= 2(m + n)$$

Since $m + n$ is an integer,

$a + b$ is divisible by 2.

Therefore,

$a + b$ is even.

Hence proved.

Example 2

Theorem

The sum of two odd integers is even.

Proof

Let

$$a = 2m + 1$$

$$b = 2n + 1$$

Then

$$a + b$$

$$= (2m + 1) + (2n + 1)$$

$$= 2m + 2n + 2$$

$$= 2(m + n + 1)$$

Since $m + n + 1$ is an integer,

$a + b$ is even.

Hence proved.

Example 3

Theorem

If n is divisible by 6, then n is divisible by 3.

Proof

Suppose

$$n = 6k$$

for some integer k .

Then

$$n = 3(2k)$$

Since $2k$ is an integer,

n is divisible by 3.

Hence proved.

5.4 Proof by Contrapositive

Definition

To prove

If p , then q

we instead prove

If not q , then not p .

This is possible because a statement and its contrapositive are logically equivalent.

General Structure

To prove:

$p \rightarrow q$

Prove:

$\neg q \rightarrow \neg p$

Example 1

Theorem

If n^2 is odd, then n is odd.

Proof

We prove the contrapositive.

Assume n is even.

Then

$n = 2k$

for some integer k .

Therefore

$$n^2 = (2k)^2$$

$$= 4k^2$$

$$= 2(2k^2)$$

which is even.

Hence if n is even, n^2 is even.

Therefore,

if n^2 is odd, n must be odd.

Hence proved.

Example 2

Theorem

If $3n + 2$ is odd, then n is odd.

Proof

Assume n is even.

Then

$$n = 2k$$

Substituting,

$$3n + 2$$

$$= 3(2k) + 2$$

$$= 6k + 2$$

$$= 2(3k + 1)$$

which is even.

Hence if $3n + 2$ is odd,

n must be odd.

5.5 Proof by Contradiction

Definition

In contradiction, we assume the statement to be false and derive an impossibility.

Since the assumption leads to a contradiction, it must be false.

Therefore the original statement is true.

General Structure

Step 1:

Assume the theorem is false.

Step 2:

Derive a contradiction.

Step 3:

Conclude the theorem is true.

Example 1

Theorem

$\sqrt{2}$ is irrational.

Proof

Assume $\sqrt{2}$ is rational.

Then

$$\sqrt{2} = a/b$$

where a and b are integers having no common factor.

Squaring both sides:

$$2 = a^2/b^2$$

Therefore

$$a^2 = 2b^2$$

Thus a^2 is even.

Hence a is even.

Let

$$a = 2k$$

Substituting:

$$4k^2 = 2b^2$$

Therefore

$$b^2 = 2k^2$$

Hence b is also even.

Thus both a and b are even.

This contradicts the assumption that a and b have no common factor.

Therefore $\sqrt{2}$ is irrational.

Example 2

Theorem

There are infinitely many prime numbers.

Proof

Assume there are only finitely many primes:

$p_1, p_2, \dots, p_n$

Consider

$$N = p_1 p_2 \dots p_n + 1$$

When N is divided by any listed prime,

remainder = 1.

Therefore none of the listed primes divides N .

Hence N has a new prime factor.

Contradiction.

Therefore infinitely many primes exist.

5.6 Proof by Cases

Definition

Sometimes a statement naturally divides into several possible cases.

If the statement is true in every case, it is true in general.

Example 1

Theorem

For every integer n ,

$n(n + 1)$ is even.

Proof

Case 1:

n is even.

Then $n(n + 1)$ is even.

Case 2:

n is odd.

Then $n + 1$ is even.

Again $n(n + 1)$ is even.

Therefore true in all cases.

Example 2

Theorem

$|n| \geq 0$ for every integer n .

Proof

Case 1:

$n \geq 0$

Then

$|n| = n \geq 0$

Case 2:

$n < 0$

Then

$|n| = -n > 0$

Therefore

$|n| \geq 0$

5.7 Biconditional Proof

Definition

To prove

$$p \leftrightarrow q$$

we prove both:

$$p \rightarrow q$$

and

$$q \rightarrow p$$

Example

Theorem

An integer n is odd if and only if n^2 is odd.

Part 1

Assume n is odd.

Then

$$n = 2k + 1$$

Squaring:

$$n^2 = (2k + 1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1$$

Hence n^2 is odd.

Part 2

Assume n^2 is odd.

Using proof by contrapositive,

if n were even,

n^2 would be even.

Contradiction.

Hence n is odd.

Therefore

n is odd iff n^2 is odd.

5.8 Existence Proofs

Definition

An existence proof demonstrates that at least one object satisfying a property exists.

Example

There exists an even prime number.

Proof:

2 is prime and even.

Therefore an even prime number exists.

Example

There exists a real number x such that

$$x^2 = 25.$$

Proof:

$x = 5$ satisfies the condition.

5.9 Uniqueness Proofs

Definition

A uniqueness proof shows that exactly one object satisfies a property.

Example

The equation

$$x + 3 = 5$$

has a unique solution.

Proof:

$x = 2$ satisfies the equation.

Suppose y is another solution.

Then

$$y + 3 = 5$$

Therefore

$$y = 2$$

Hence $y = x$.

Therefore the solution is unique.

5.10 Common Errors in Proofs

A proof is valid only if every step is logically justified.

False Proof That $1 = 2$

Assume

$$a = b$$

Then

$$a^2 = ab$$

Subtract b^2 :

$$a^2 - b^2 = ab - b^2$$

Factor:

$$(a - b)(a + b) = b(a - b)$$

Divide by $(a - b)$:

$$a + b = b$$

Since $a = b$,

$$2b = b$$

Therefore

$$2 = 1$$

Error

Since

$$a = b$$

we have

$$a - b = 0$$

The proof divides by zero.

Division by zero is undefined.

Hence the argument is invalid.

5.11 Mathematical Induction

Introduction

Mathematical induction is a proof technique used to establish statements involving positive integers.

It is based on the principle:

"If the first domino falls and each domino knocks down the next one, then all dominos fall."

Principle of Mathematical Induction

To prove $P(n)$ for all $n \geq 1$:

Step 1: Basis Step

Show $P(1)$ is true.

Step 2: Induction Hypothesis

Assume $P(k)$ is true.

Step 3: Inductive Step

Show $P(k+1)$ is true.

Then $P(n)$ is true for all positive integers.

Example 1

Prove

$$1 + 2 + 3 + \dots + n$$

$$= n(n+1)/2$$

Basis Step

For $n = 1$

$$\text{LHS} = 1$$

$$\text{RHS} = 1(2)/2 = 1$$

True.

Induction Hypothesis

Assume

$$1 + 2 + \dots + k$$

$$= k(k+1)/2$$

Inductive Step

$$1 + 2 + \dots + k + (k+1)$$

$$= k(k+1)/2 + (k+1)$$

$$= (k+1)(k+2)/2$$

Hence true for $k+1$.

Therefore proved.

Example 2

Prove

$$n < 2^n$$

for all positive integers n .

Basis Step

$$1 < 2$$

True.

Inductive Step

Assume

$$k < 2^k$$

Then

$$k+1 \leq 2k < 2 \cdot 2^k = 2^{k+1}$$

Hence true.

Example 3

Prove

$$n^3 - n$$

is divisible by 3.

Basis Step

$$1^3 - 1 = 0$$

Divisible by 3.

Inductive Step

$$(k+1)^3 - (k+1)$$

$$= (k^3 - k) + 3(k^2 + k)$$

Both terms are divisible by 3.

Hence proved.

5.12 Strong Mathematical Induction

In ordinary induction we assume only $P(k)$.

In strong induction we assume:

$P(1), P(2), \dots, P(k)$

are all true.

Example

Every integer greater than 1 can be expressed as a product of prime numbers.

Basis Step

2 is prime.

Inductive Hypothesis

Assume every integer from 2 to k can be written as a product of primes.

Inductive Step

Consider $k+1$.

If prime, done.

If composite,

$$k+1 = ab$$

where $a, b < k+1$.

By induction hypothesis,

a and b are products of primes.

Therefore

$k+1$ is also a product of primes.

Hence proved.

Summary

In this chapter we studied:

- Direct Proof
- Proof by Contrapositive
- Proof by Contradiction
- Proof by Cases
- Biconditional Proof
- Existence Proof
- Uniqueness Proof
- Common Errors in Proofs
- Mathematical Induction
- Strong Mathematical Induction

These proof techniques form the foundation of higher mathematics, graph theory, abstract algebra, automata theory, number theory, and theoretical computer science.